A Hindi farm assistant on the farmer's own phone – supported by village SHGs

AgriAI helps a smallholder farmer check a sick crop from a photo, know when to water, get weather warnings, and sell produce – in simple Hindi, by touch or by voice. It is built and tested with demo data. **We request a supervised 4-week pilot in Churu district with KVK and SHG support.**

The problem in Rajasthan

- Rajasthan grew **45.40%** of India's rapeseed & mustard and **14.66%** of India's nutri/coarse cereals in 2023-24 [S1]. Losses on these farms matter for the whole country.
- **219 of Rajasthan's 302** groundwater assessment units were over-exploited in the 2022 assessment [S2]. Every unnecessary irrigation counts.
- Pests and diseases destroy **up to 40%** of global crop production each year [S3]. Early, correct identification makes the difference.
- Expert advice exists – Rajasthan has **47 Krishi Vigyan Kendras** [S4] – but a farmer often needs an answer in the field, the same day, in their own language.

What AgriAI does

AI crop advisor

Photo → problem, cause, desi remedies, generic medicine names, precautions; follow-up questions; 🔊 read-aloud.

Farm advice

Weather alerts (heat, frost, rain, dry spell), water timing by crop stage, crop calendar – written rules, marked "verify with KVK".

Market & ecosystem

Sell by photo + voice, orders, delivery tracking link, shared mandi trips, machine rental, scheme information with official links.

SHG coordinator

Village SHG didi verifies sellers, helps non-readers list, resolves complaints in 48 h, runs mandi trips; pilot dashboard.

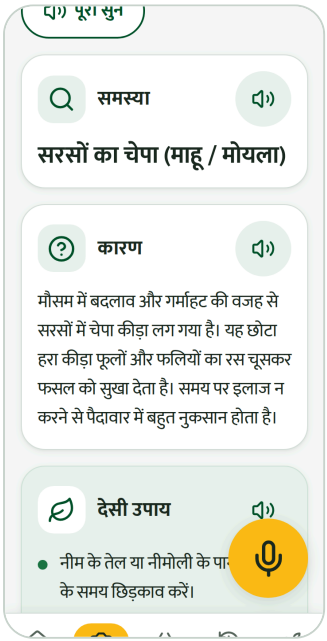
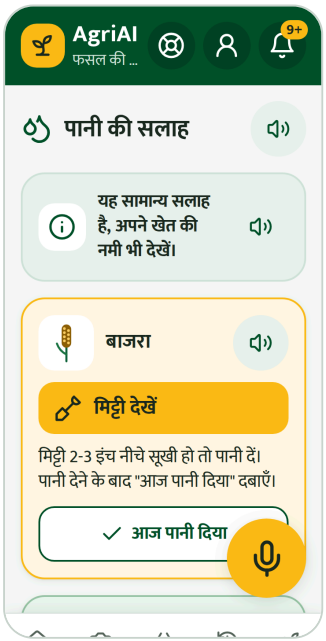
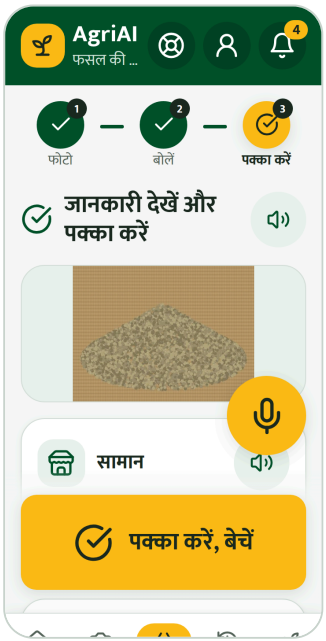


Photo check: real AI answer (recorded)



Water advice by crop stage



Selling: AI fills the form



Pilot dashboard: counts only

Safety first

- **Refuses to guess:** if the photo is unclear, not a plant, or the AI is unsure, the app asks for a new photo – no answer is shown.
- **Medicine blocklist:** banned and restricted pesticides are removed by a server check after the AI; only generic names, never an exact dose.
- **KVK line on every answer:** "confirm with your nearest Krishi Vigyan Kendra".
- **Accuracy:** not yet measured on field photos. A KVK expert review of 100 labelled photos is planned before the pilot (pass mark: ≥70% correct, 100% of unclear photos refused, 0 banned medicines).

Data privacy

- The AI receives **only the photo and the crop name** – never a name, phone number or location (automatically tested). Check photos are not stored.
- Consent screen in simple Hindi with read-aloud; **"मेरा डेटा"** lets a farmer download or delete their data at once; append-only audit log of coordinator actions; encrypted backups kept 14 days.
- Designed around India's DPDP Act, 2023 and DPDP Rules, 2025 (notified 14 Nov 2025, 18-month phased compliance) [S5]. Policy and terms are drafts awaiting legal review.

How it can support existing government efforts

- **KVKs:** every answer points farmers to the KVK; all thresholds are open for KVK experts to verify and correct.
- **SHGs (e.g. Rajeevika groups):** a defined role for SHG members as village coordinators (compensation to be agreed with the SHGs).
- **Schemes:** plain-Hindi scheme information that links to the official website; it does not process applications.
- **Public data:** can show mandi prices from data.gov.in. *No official partnership exists today.*

Honest status

Built and working	Photo check, chat, voice, weather/water/calendar, marketplace, delivery link, complaints, mandi trips, rentals, schemes, coordinator panel, consent and "मेरा डेटा", offline demo mode.
Tested with demo data	566 server tests, 146 app tests, 8 end-to-end phone journeys incl. accessibility; real AI answers recorded on openly licensed disease photos.
Planned / needs pilot	KVK accuracy review, cloud deployment, legal review, real farmers and SHGs, effect on yield or income (not claimed).

Running cost (estimate)

Hosting (Render Starter)	**$7 / month**	[S10]
Database (MongoDB Atlas)	**$0 (free, 512 MB) – $8 / month**	[S8]
AI (Gemini 3.8 Flash)	**≈ $8 per 1,000 photo checks***	[S6]
Pilot month, 50 farmers	**≈ $15–32 (≈ ₹1,300–2,800)***	

*Estimate: ≈3,500 input + 1,500 output tokens per check; price until 31 Dec 2026 (doubles from 1 Jan 2027); a free tier may cover the pilot; ₹88/$ assumed. Render's free tier is not for production [S7].

Our request

1. Permission for a **supervised 4-week pilot in Churu district** with 20–50 farmers and 2–3 SHG groups.
2. **Time of one KVK expert (Churu):** review 100 photos before the pilot and sample answers during it.
3. An introduction to **2–3 SHG groups** willing to act as coordinators.
4. We share weekly results and pre-agreed stop/continue rules with the district administration. **No funds requested for the pilot.**

Sources: [S1] Economic Survey 2023-24, Stat. Appendix Table 1.18 · [S2] PIB, Ministry of Jal Shakti, 8 Dec 2022 · [S3] FAO, 2 Jun 2021 · [S4] ICAR KVK list · [S5] PIB, MeitY, 14 Nov 2025 · [S6] Gemini API pricing · [S7] Render docs · [S8] MongoDB Atlas pricing · [S10] Render, May 2026. Links: showcase/SOURCES.md

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